

Aditya Indurkhya

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EDUCATION

M.S., Data Analytics | George Mason University, Fairfax, VA | GPA: 3.73/4.0

Aug 2024 - May 2026

B.Tech., Computer Science & Engineering | Gyan Ganga Institute of Technology, India

Aug 2020 - June 2024

PROFESSIONAL EXPERIENCE

Data Engineer | DMV Music Alliance | Washington, D.C.

Aug 2026 - Present

- Architected ETL pipelines consolidating 3 disparate venue, artist, and event feeds into a unified schema with validation and deduplication, eliminating 72 hours of manual data entry weekly.
- Automated recurring data workflows across Portal Development and Data teams, replacing spreadsheet updates with scheduled weekly refreshes and reducing stale portal listings 80%.
- Shipped responsive booking and discovery pages for artist profiles, venue listings, and event search, supporting 720+ artist and venue profiles at platform launch.

Data Engineer | ICSS | India | IT hardware supplier, government procurement

Jul 2022 - Aug 2023

- Owned the company's first unified data warehouse, engineering PySpark ETL pipelines across 8 heterogeneous sources and 12M+ monthly records, cutting daily runtime 30% while maintaining 95%+ uptime.
- Reduced p95 query latency 49% (35s to 18s) across 50GB+ datasets by redesigning PostgreSQL indexes and optimizing hot-path SQL across PostgreSQL and Redshift, accelerating reporting from same-afternoon to same-morning.
- Instrumented Python-based data validation and anomaly alerting, reducing pipeline issues from 12% to 5% of daily runs, saving 15 staff-hours weekly and establishing 5 governance standards team-wide.

Data Engineering Intern | D'Festo Private Limited | Jabalpur, India

May 2021 - May 2022

- Established the company's first end-to-end ETL and reporting system using Python, Pandas, and PostgreSQL, processing 200K+ records daily across 3 source systems with validation and error handling.
- Resolved 20 data quality defects involving duplicates, nulls, and key violations through end-to-end reporting audits, reducing recurring query times 40% through targeted PostgreSQL indexing.
- Produced 4 Power BI dashboards with DAX-based KPIs, replacing manual spreadsheet tracking, and authored runbooks enabling non-engineers to triage ingestion alerts without engineering escalation.

PROJECTS

Lakehouse Copilot – Streaming Lakehouse with Natural Language Querying | Kafka, PySpark, Iceberg, dbt, Airflow, Trino, pgvector

- Engineered a streaming lakehouse ingesting live events through Kafka into Apache Iceberg tables on S3 via PySpark Structured Streaming, with dbt star-schema transforms orchestrated through Airflow.
- Grounded a text-to-SQL agent in dbt-generated schema metadata using pgvector retrieval; benchmarked on a 100-question golden set with execution accuracy enforced as a GitHub Actions CI gate.
- Exposed governed data marts through an MCP server with SQL validation, row limits, and read-only guardrails; provisioned S3, Glue Data Catalog, and Athena infrastructure with Terraform.

DataAgent Pro – Stateful Multi-Agent Data Analysis System | LangGraph, LangChain, FastAPI, React, DuckDB, PostgreSQL

- Orchestrated a LangGraph multi-agent workflow routing SQL, Python, visualization, and reporting tasks across 10+ specialized nodes, with PostgreSQL checkpointing for persistent state.
- Hardened execution with SQL validation, Python AST security checks, and sandboxed subprocesses feeding a reflection-retry loop to prevent incorrect results from reaching reports.
- Deployed a full-stack platform with React, FastAPI, and DuckDB, integrating MCP tools, Plotly visualizations, PDF reporting, and observability for latency and retry rates.

ClimateGPT – MCP Server for Climate Risk Intelligence | FastAPI, PostgreSQL, pgvector, GeoPandas, Docker

- Developed a FastAPI MCP server spanning 180+ countries and 16 WRI Aqueduct indicators, routing natural-language queries through structured SQL retrieval and pgvector semantic search.
- Designed a dual-schema ETL pipeline processing Aqueduct v4.0 shapefile, Excel, and PDF sources into core and vector PostgreSQL schemas supporting RCP 4.5/8.5 scenario comparisons.
- Benchmarked against WRI Aqueduct ground truth using containerized pytest and per-indicator scoring, achieving 70% accuracy through Docker Compose and GitHub Actions CI.

DataPulse AI – Data Quality & Observability Platform | Python, SQLite, Streamlit, Plotly, Docker

- Prototyped an observability pipeline across 5,000+ retail records with scheduled quality checks, detecting 100% of deliberately injected data failures.
- Implemented volume, schema, drift, and lineage monitoring with statistical anomaly detection and DFS-based column-level lineage to surface downstream impact.
- Delivered a 6-tab Streamlit dashboard covering quality, anomalies, schema, lineage, and incidents, with automated root-cause and remediation reports.

TECHNICAL SKILLS

Languages: Python, SQL, R

Data Engineering: ETL/ELT, Data Warehousing, Dimensional Modeling, Query Optimization, Data Quality & Governance, Data Lineage

Data Processing & Orchestration: Apache Spark (PySpark), Apache Kafka, Apache Airflow, dbt, Apache Iceberg, Pandas

Databases & Warehouses: PostgreSQL (PostGIS, pgvector), Amazon Redshift, Snowflake, Databricks, Trino, DuckDB, SQLite

Cloud & Engineering Tools: AWS (S3, Glue, Athena), Terraform, Docker, FastAPI, REST APIs, Git/GitHub, GitHub Actions, CI/CD, pytest

AI/LLM & Analytics: RAG, LangGraph, LangChain, MCP, Multi-Agent Systems, Claude API, LLM Evaluation, ARIMA, A/B Testing, Power BI (DAX), Tableau, Streamlit, Plotly

CERTIFICATIONS

AWS Academy Data Engineering | SQL for Data Analysis (LinkedIn) | dplyr and tidymodels (DataCamp)